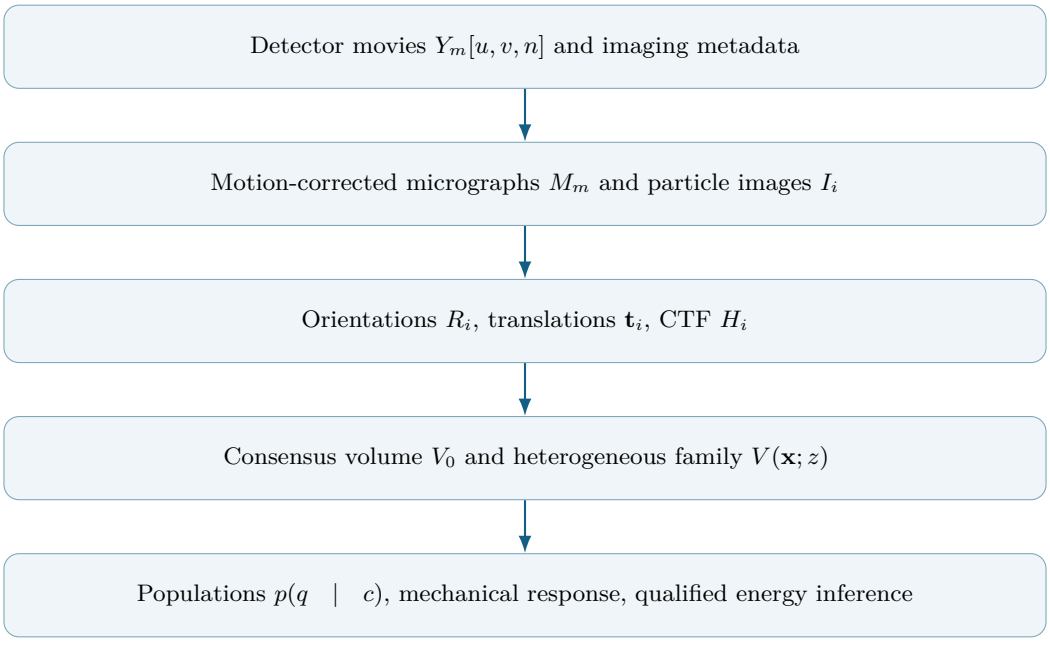
The Mathematics of Single-Particle Cryo-EM

A proof-oriented exposition with numerical examples, visualization, and physically interpretable inference

Expanded technical study edition

From detector movie arrays to Fourier reconstruction, pose marginalization, continuous heterogeneity, neural models, free-energy inference, and force-dependent PP2A mechanics

Main mathematical problem. Single-particle cryo-EM produces many noisy two-dimensional images of copies of a three-dimensional molecule in unknown orientations and, often, different conformations. The inverse problem is to infer a 3D structural model, particle-by-particle poses, and (where supported) a distribution of conformational states. A mechanical perturbation such as a DNA spring creates an additional statistical-mechanical inverse problem: determine whether the observed structural populations change in a way consistent with a calibrated mechanical bias, rather than pose errors or selection bias.



October 2026 | Research-study companion for PP2A A-subunit / DNA-spring work

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1 How to read this exposition: mathematical objects, assumptions, and the task

A cryo-EM pipeline manipulates several mathematically different kinds of data. They should never be conflated. An electron detector movie is a stack of pixel arrays; a particle crop is a single two-dimensional image; a reconstructed map is a three-dimensional scalar field; an atomic model is a collection of coordinates and chemical identities; and a conformational ensemble is a probability distribution over structures. These are distinct outputs with distinct uncertainties.

Object	Exact mathematical form	Typical units
Raw movie	$Y_m \in \mathbb{R}^{H\times W\times T}$	counts / pixel
Micrograph	$M_m \in \mathbb{R}^{H\times W}$	calibrated intensity
Particle image	$I_i \in \mathbb{R}^{D\times D}$	processed intensity
2D Fourier image	$\widehat{I}_i \in \mathbb{C}^{D\times D}$	convention-dependent
Volume grid	$V \in \mathbb{R}^{D_v\times D_v\times D_v}$	potential / voxel
Particle rotation	$R_i \in SO(3)$	dimensionless
Particle shift	$\mathbf{t}_i \in \mathbb{R}^2$	pixels or Å
Latent variable	$z_i \in \mathbb{R}^d$	arbitrary unless calibrated
Physical coordinate	$q_i \in \mathbb{R}^p$	Å, radians, etc.
Atomic model	$\{(\mathbf{r}_j, s_j, B_j, o_j)\}$	Å, atomic parameters

The symbols m, i, j index movies, particle images, and atoms, respectively. u, v, n index detector pixels and movie frames. H, W, T, D, D_v are positive integers denoting array dimensions. N is the number of extracted particles. $\mathbf{x} = (x, y, z)$ is a real-space 3D coordinate; $\mathbf{k} = (k_x, k_y)$ is a 2D spatial frequency; $\mathbf{K} = (K_x, K_y, K_z)$ is a 3D spatial frequency. A hat indicates a Fourier transform; $\mathbb{C}$ is the field of complex numbers. "Ground truth" means a quantity known in a synthetic benchmark, not one estimated from real cryo-EM data.

Four separate inverse problems. (1) *Imaging:* remove detector/motion and estimate CTF. (2) *Geometry:* infer viewing directions and shifts. (3) *Structure:* infer one or many 3D maps. (4) *Physics:* infer probabilities and mechanical response, which requires additional assumptions beyond image reconstruction.

2 From an electron detector movie to a particle image

2.1 The detector records a discrete stochastic field

A continuous image is a function $I : \mathbb{R}^2 \to \mathbb{R}$. The detector instead records integers or calibrated real values at discrete pixels, which we denote $Y_m[u, v, n]$. At the simplest level, electron counting is random. If N_{uvn} is the number of detected events in a frame, a model is

$$N_{uvn} \sim \text{Poisson}(\lambda_{uvn}), \qquad p(N = n \mid \lambda) = e^{-\lambda}\frac{\lambda^n}{n!}. \tag{1}$$

λ_{uvn} is the expected count at a given pixel/frame and n is a nonnegative integer. For a Poisson distribution, $\mathbb{E}[N] = \text{Var}(N) = \lambda$. Real electron detectors introduce gain normalization, counting corrections, correlated noise, and other details; downstream models often approximate the corrected noise as Gaussian. This approximation must be distinguished from the underlying detector statistics.

Worked example

Suppose a pixel has an expected count of $\lambda = 4$ during one frame. The probability of observing exactly 2 events is

$$p(N = 2 \mid 4) = e^{-4}\frac{4^2}{2!} = 8e^{-4} \approx 0.1465.$$

The standard deviation is $\sqrt{4} = 2$ events, so the single-pixel relative noise is large. This is why class averaging and the combination of many particles are indispensable.

2.2 Pixel coordinates, field of view, and Nyquist resolution

For a $D \times D$ extracted image with calibrated spacing a in Å/pixel, the physical coordinates relative to its center are

$$\mathbf{x}_{uv} = a \begin{pmatrix} u - (D-1)/2 \\ v - (D-1)/2 \end{pmatrix}, \quad u, v \in \{0, \dots, D-1\}. \tag{2}$$

The field-of-view width is approximately Da. On a discrete grid the largest representable spatial frequency along an axis is $k_{\mathrm{Nyq}} = 1/(2a)$ cycles per Å; equivalently, the Nyquist spatial period is $d_{\mathrm{Nyq}} = 2a$. This is a sampling limit, *not* a demonstrated experimental map resolution.

Worked example

At $D = 256$ and $a = 1.25$ Å/pixel, a particle box spans $256(1.25) = 320$ Å. Its sampling Nyquist resolution is $2(1.25) = 2.5$ Å. A resulting reconstruction may be much worse than 2.5 Å because of noise, flexibility, CTF information loss, or preferred orientation. If a 300 Å-wide DNA construct must fit in the box, centering and border margins become important.

2.3 Frame alignment and dose weighting

Model an aligned object g moving from frame to frame by unknown translations $\boldsymbol{\delta}_n$:

$$Y_n(\mathbf{x}) = g(\mathbf{x} - \boldsymbol{\delta}_n) + \epsilon_n(\mathbf{x}). \tag{3}$$

ϵ_n is the frame's noise, and $\boldsymbol{\delta}_n \in \mathbb{R}^2$ the in-plane displacement. Under Gaussian pixel noise, a least-squares registration criterion is

$$\min_{g, \{\boldsymbol{\delta}_n\}} \sum_n \| Y_n(\cdot + \boldsymbol{\delta}_n) - g \|_2^2 . \tag{4}$$

Once the shifts are estimated, a weighted sum creates a corrected micrograph. More accurately, radiation damage motivates frequency-dependent weights:

$$\widehat{M}(\mathbf{k}) = \frac{\sum_{n=1}^{T} w_n(\mathbf{k}) e^{2\pi \mathrm{i} \mathbf{k} \cdot \widehat{\boldsymbol{\delta}}_n} \widehat{Y}_n(\mathbf{k})}{\sum_{n=1}^{T} w_n(\mathbf{k})}. \tag{5}$$

$w_n(\mathbf{k}) \geq 0$ is the weight for frame n at frequency $\mathbf{k}$. This expression describes the idea of aligning and combining frames, not the exact implementation of a particular motion-correction package. Patch motion correction replaces one displacement per frame with a smoothly varying field of local shifts.

Worked example

Suppose the same sharp feature is displaced by 2 pixels in frame 2. Averaging frames without aligning them creates a ghost or blur. If the Fourier frequency is $k_x = 1/8$ cycles/pixel, the 2-pixel shift introduces phase $e^{-2\pi \mathrm{i}(1/8)2} = e^{-\pi \mathrm{i}/2} = -\mathrm{i}$. Multiplication by the inverse phase $+\mathrm{i}$ aligns that Fourier component. This is the Fourier shift theorem in action.

2.4 Picking and extracting a particle

A picker returns candidate centers $\mathbf{c}_i = (c_{x,i}, c_{y,i})$. Extraction forms

$$I_i[u, v] = M_m[c_{x,i} + u,\ c_{y,i} + v], \tag{6}$$

where u, v are offsets within the crop. When the center is not an integer pixel position, interpolation is required. Particle metadata must retain the originating micrograph, location, pixel calibration, and CTF. A crop is not yet an orientation estimate; it is only a sampled 2D observation.

2.5 Fourier cropping and aliasing

Downsampling by factor r increases the physical pixel size from a to ra. To avoid aliasing, remove frequencies above the *new* Nyquist frequency before decimation. Otherwise different original frequencies collapse onto the same sampled frequency. Fourier cropping is therefore a deliberate low-pass operation; it cannot preserve high-resolution information beyond the new Nyquist limit.

PP2A-specific caution. For the spring construct, a box centered tightly around A-subunit density may omit most of the DNA arc. A larger field captures spring geometry but consumes more GPU memory and can allow the appendage to dominate alignment. Using linked large/low-resolution and small/high-resolution crops is a modeling choice, not free extra independent data; the two crops share noise and physical signal.

3 The 3D molecular object and its imaging geometry

3.1 Potential map versus atomic coordinates

A cryo-EM 3D reconstruction is an estimate of an electrostatic scattering-potential map $V(\mathbf{x})$, often informally called "density." The corresponding discretized array has D_v^3 voxels. It is distinct from an X-ray crystallographic electron-density map. An atomistic model is a set of coordinates and species $\{(\mathbf{r}_j, s_j)\}_{j=1}^M$ plus occupancies and displacement parameters. To illustrate how atomic structure can generate a smooth volume, write the deliberately approximate sum

$$V(\mathbf{x}) \approx \sum_{j=1}^{M} A_j \exp\left[-\frac{\|\mathbf{x}-\mathbf{r}_j\|^2}{2\sigma_j^2}\right]. \tag{7}$$

Here A_j is the contribution amplitude, σ_j a blurring scale, and $\|\mathbf{x}-\mathbf{r}_j\|$ Euclidean separation. Accurate calculation replaces this toy Gaussian form with physical electron scattering factors. The image inverse problem first reconstructs a map; fitting atomic coordinates is a separate step.

3.2 Rotation matrices and $SO(3)$

The molecular orientation is $R \in SO(3)$, the set of 3×3 real matrices satisfying $R^\mathsf{T}R = I$ and $\det R = 1$. Here I is the identity, R^T the transpose, and the determinant condition excludes reflections. A rotation through angle θ about the electron-beam direction is

$$R_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{8}$$

Three parameters are required for generic orientation, often Euler angles or quaternions. Euler angles have singularities and convention choices; a normalized quaternion offers a convenient numerical representation. We define R throughout to map *molecular-frame* coordinates to *laboratory-frame* coordinates. This fixes the otherwise confusing transpose convention in the projection formula.

3.3 Deriving the projection operator

In a weak-scattering, projection approximation the ideal projected potential for a beam parallel to laboratory z is

$$(PV)(x,y) = \int_{-\infty}^{\infty} V(x,y,z)\,\mathrm{d}z. \tag{9}$$

P maps a 3D function to a 2D function by integrating along the beam. For a rotated molecule, evaluate the reference molecular map at $R^\mathsf{T}(x,y,z)^\mathsf{T}$:

$$(P_R V)(x,y) = \int_{-\infty}^{\infty} V\left(R^\mathsf{T}\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right)\mathrm{d}z. \tag{10}$$

R^{T} appears because its inverse maps laboratory positions into the reference molecular frame. The idealized imaging model omits multiple scattering, Ewald-sphere curvature, and other higher-order effects; those can matter when the simplifying approximation is inadequate.

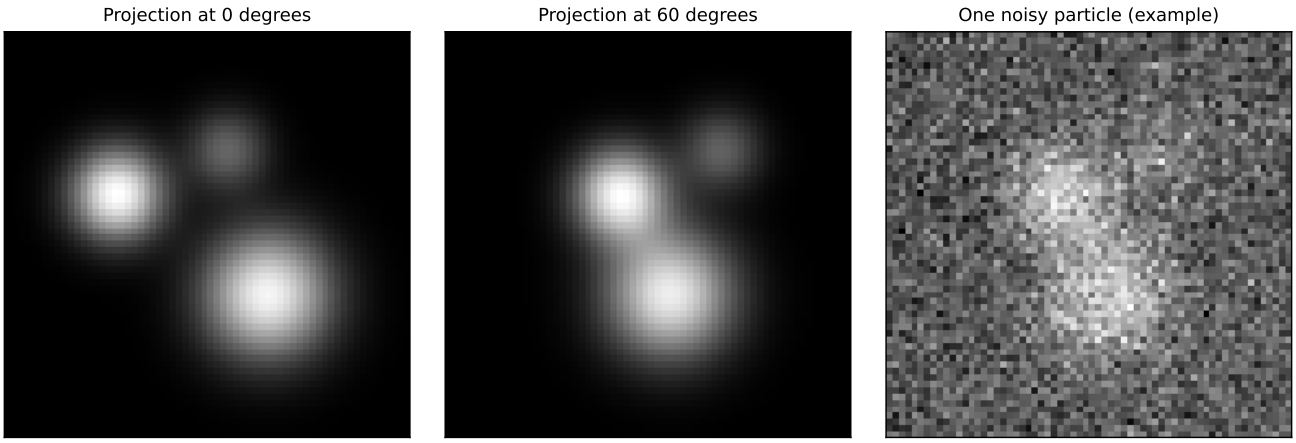


Figure 1: An asymmetric synthetic 3D volume projected from two viewing directions; the third panel adds strong Gaussian noise to one projection. These are generated teaching images, not experimental particles.

Worked example

A toy volume has four depth layers of thickness 1 Å with potential values $(1, 3, 4, 2)$ in arbitrary units along a given ray. The projected value is approximately $1(1) + 3(1) + 4(1) + 2(1) = 10$ potential $\times$ Å. Every 3D depth sequence with the same sum gives the same projection at that pixel. This explicitly proves that a single view cannot recover depth ordering without additional constraints.

3.4 Why many rotated particles solve a tomography problem

Worked example

A single projection collapses depth. A differently oriented particle produces a different linear combination of the internal values. In a simplified two-voxel toy object, a first measurement might be $y_1 = v_1 + v_2$ and a second $y_2 = v_1 - v_2$. Then

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \quad v_1 = \frac{y_1 + y_2}{2}, \quad v_2 = \frac{y_1 - y_2}{2}.$$

If $(y_1, y_2) = (8, 2)$, the recovered values are $(5, 3)$. Real tomography involves millions of variables, irregular angular sampling and a CTF, but its identifiability logic is the same: complementary measurements disentangle unknown 3D values.

4 Fourier analysis: the central-slice theorem and its consequences

4.1 Fourier conventions and dimensions

Define the 3D Fourier transform (using spatial frequency in cycles per length) by

$$\widehat{V}(\mathbf{K}) = \int_{\mathbb{R}^3} V(\mathbf{x}) e^{-2\pi \mathrm{i} \mathbf{K}\cdot\mathbf{x}} \, \mathrm{d}^3\mathbf{x}, \qquad V(\mathbf{x}) = \int_{\mathbb{R}^3} \widehat{V}(\mathbf{K}) e^{2\pi \mathrm{i} \mathbf{K}\cdot\mathbf{x}} \, \mathrm{d}^3\mathbf{K}. \tag{11}$$

$\mathbf{K}$ has units of reciprocal length; $\mathrm{i}^2 = -1$ and $\mathbf{K} \cdot \mathbf{x} = K_x x + K_y y + K_z z$. Frequency $\|\mathbf{K}\| = 1/d$ corresponds to structural period d. Similar definitions hold in 2D. The discrete Fourier transform uses finite sums and normalization conventions; the mathematical reasoning is unchanged.

4.2 Proof of the Fourier-slice theorem

For unrotated projection $g(x, y) = \int V(x, y, z)\, \mathrm{d}z$, take its 2D transform:

$$\widehat{g}(k_x, k_y) = \int_{\mathbb{R}^2} g(x, y) e^{-2\pi \mathrm{i}(k_x x + k_y y)}\, \mathrm{d}x\, \mathrm{d}y \tag{12}$$

$$= \int_{\mathbb{R}^2} \int_{\mathbb{R}} V(x, y, z) e^{-2\pi \mathrm{i}(k_x x + k_y y)}\, \mathrm{d}z\, \mathrm{d}x\, \mathrm{d}y \tag{13}$$

$$= \int_{\mathbb{R}^3} V(x, y, z) e^{-2\pi \mathrm{i}(k_x x + k_y y + 0z)}\, \mathrm{d}^3\mathbf{x} \tag{14}$$

$$= \widehat{V}(k_x, k_y, 0). \tag{15}$$

The first equality is the definition, the second substitutes the projection, the third combines integrals under suitable integrability assumptions, and the last recognizes the 3D transform at $K_z = 0$. For rotated objects,

$$\boxed{\mathcal{F}_2[P_R V](k_x, k_y) = \widehat{V}\left(R^{\mathsf{T}} \begin{pmatrix} k_x \\ k_y \\ 0 \end{pmatrix} \right).} \tag{16}$$

This is the *central-slice theorem*: each 2D Fourier image samples a central plane through the 3D Fourier volume. Rotation changes the plane orientation. Reconstruction combines these planes (with proper weights) before taking an inverse 3D Fourier transform.

4.3 Two related theorems: shifts and convolution

If $T_{\mathbf{t}} g(\mathbf{x}) = g(\mathbf{x} - \mathbf{t})$, then

$$\mathcal{F}[T_{\mathbf{t}} g](\mathbf{k}) = e^{-2\pi \mathrm{i} \mathbf{k} \cdot \mathbf{t}} \widehat{g}(\mathbf{k}). \tag{17}$$

A translation becomes multiplication by a unit-magnitude complex phase. If $h * g$ is a spatial convolution,

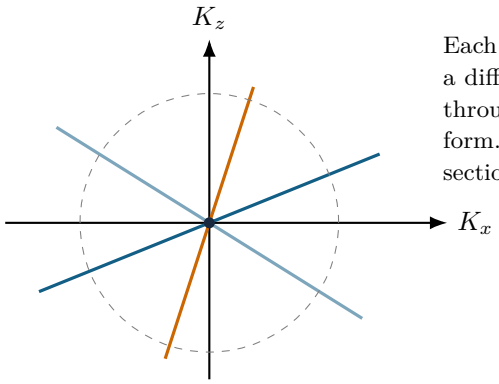
$$\mathcal{F}[h * g](\mathbf{k}) = \widehat{h}(\mathbf{k}) \widehat{g}(\mathbf{k}). \tag{18}$$

These two identities explain the compact Fourier-domain representation of particle translations and microscope transfer functions.

Worked example

Take an image Fourier coefficient at $k_x = 0.1$ cycles/pixel, $k_y = 0$, and translate the particle rightward by $t_x = 2$ pixels. The multiplicative factor is $e^{-2\pi \mathrm{i}(0.1)(2)} = e^{-0.4\pi \mathrm{i}} = \cos(0.4\pi) - \mathrm{i} \sin(0.4\pi) \approx 0.309 - 0.951\mathrm{i}$. Its magnitude is 1, so translation changes the *phase* but not the magnitude of that Fourier coefficient.

4.4 A visualization of Fourier-plane sampling



Each rotated particle contributes a differently oriented planar slice through the same 3D Fourier transform. This 2D drawing is a cross-section of that geometry.

Figure 2: Central Fourier slices from different particle orientations; schematic, not real data.

5 CTF physics and the full measurement model

5.1 Where image contrast comes from

A high-energy electron wave interacting with a sufficiently weak, thin specimen can be approximated as an exit wave $\psi_{\text{exit}}(x, y) \approx 1 + \mathrm{i}\varphi(x, y)$, where φ is a small projected phase shift proportional, in a suitable convention, to the electrostatic potential integrated along the beam. Without phase contrast, such a weak phase object would be difficult to image. Defocus and lens aberrations convert exit-wave phase variations into recorded intensity contrast. In Fourier space this is modeled by a contrast transfer function (CTF), which modulates the signed contrast at each spatial frequency.

5.2 CTF as a frequency filter

A common sign convention for the weak-phase CTF is

$$H(k) = -\left[\sqrt{1 - A^2}\sin\chi(k) + A\cos\chi(k)\right], \qquad \chi(k) = \pi\lambda\Delta f\, k^2 - \frac{\pi}{2}C_s\lambda^3 k^4 + \chi_0. \tag{19}$$

$k = \|\mathbf{k}\|$ is the frequency radius (cycles/Å); A is the amplitude-contrast fraction; λ is electron wavelength (Å); Δf defocus (Å); C_s spherical aberration coefficient (Å); χ_0 additional phase shift. This is an illustrative isotropic model; astigmatism makes defocus depend on azimuth, and real imaging includes envelopes and other aberrations. The signs of defocus and CTF are software-convention dependent.

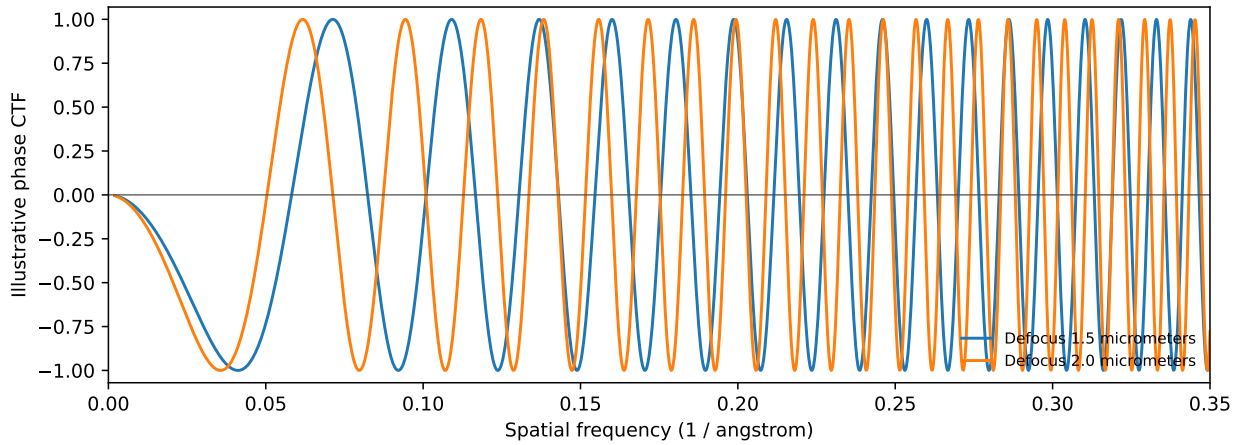


Figure 3: Illustrative isotropic phase CTF curves for two defocus values (electron wavelength approximately 0.0197 Å, spherical aberration 2.7 mm). Distinct CTF zeros motivate combining observations from different defocus conditions.

Frequency-by-frequency interpretation. If $H(\mathbf{k}) = +0.8$, the molecular coefficient at that frequency is scaled by 0.8. If $H = -0.8$, the contrast is inverted. If $H = 0$, that frequency's molecular signal is absent from that observation. Hence different defocus settings and many particles are needed to cover missing information. Dividing blindly by H near zeros amplifies noise enormously.

5.3 Wiener-like frequency reconstruction

Suppose one Fourier coefficient v generates noisy measurements $y_i = h_i v + \epsilon_i$, where h_i are known CTF values and independent noise has variance σ_i^2. Minimize weighted residuals plus a Gaussian-prior penalty $\lambda|v|^2$:

$$L(v) = \sum_i \frac{|y_i - h_i v|^2}{\sigma_i^2} + \lambda |v|^2.$$

Differentiating with respect to the real and imaginary components of v, or using complex least squares, gives

$$\boxed{\widehat{v} = \frac{\sum_i h_i^* y_i / \sigma_i^2}{\sum_i |h_i|^2 / \sigma_i^2 + \lambda}.} \tag{20}$$

The numerator coherently combines corrected observations, and the denominator discounts frequencies with low CTF transmission. h_i^* is the complex conjugate (the CTF is often effectively real under standard assumptions). When $h_i = 0$, observation i contributes no information to that coefficient.

Worked example

Take one true Fourier coefficient $v = 10$, and two noiseless observations with CTFs $h_1 = 0.8$ and $h_2 = -0.6$. The detector observes $y_1 = 8$ and $y_2 = -6$. With equal variance and $\lambda = 0$,

$$\widehat{v} = \frac{0.8(8) + (-0.6)(-6)}{0.8^2 + (-0.6)^2} = \frac{6.4 + 3.6}{0.64 + 0.36} = 10.$$

Simply averaging raw values would yield $(8 - 6)/2 = 1$, catastrophically cancelling the correct signal. The sign of the CTF is essential.

5.4 The per-particle cryo-EM forward equation

Under the projection approximation, a useful complex Fourier-domain equation is

$$\boxed{\widehat{I}_i(\mathbf{k}) = H_i(\mathbf{k}) e^{-2\pi i \mathbf{k}\cdot\mathbf{t}_i} \widehat{V}\left(R_i^{\mathsf{T}} \begin{pmatrix} k_x \\ k_y \\ 0 \end{pmatrix}; z_i\right) + \widehat{\epsilon}_i(\mathbf{k}).} \tag{21}$$

- **Inputs to the forward simulation:** 3D volume $V(\cdot; z_i)$, orientation R_i, 2D shift $\mathbf{t}_i$, CTF H_i, pixel grid and noise statistics.
- **Output:** the predicted complex Fourier image $\widehat{I}_i$, and after inverse 2D DFT, its real-space particle image.
- **Inverse task:** images and imaging metadata are observed; volume, poses and usually conformation are hidden. The fitted model must explain all images simultaneously.

For Gaussian Fourier noise with covariance Σ_i, the log likelihood is proportional to

$$-2\log p(\widehat{I}_i \mid V, R_i, \mathbf{t}_i, z_i) = (\widehat{I}_i - \widehat{I}_i^{\text{pred}})^* \Sigma_i^{-1} (\widehat{I}_i - \widehat{I}_i^{\text{pred}}) + \text{constant}. \tag{22}$$

A star on a vector denotes conjugate transpose. This is a quadratic *Mahalanobis distance*. If $\Sigma_i = \sigma^2 I$, it reduces to a scaled sum of squared residuals. Processed particle images generally have nonuniform noise; whitening attempts to transform the residuals so their covariance is closer to the identity.

6 Probability theory for alignments, averages, and classes

6.1 Why one image cannot determine one pose exactly

Low-dose electron microscopy is noisy. Even with a reference volume, several projections may fit a single image nearly equally well. Consequently, poses are uncertain random variables; an orientation assignment is an estimate, not a direct observation. Define a particle's hidden state $h_i = (R_i, \mathbf{t}_i, c_i)$, where c_i may index a structural class. Bayes' rule gives

$$p(h_i \mid I_i, V) \propto p(I_i \mid h_i, V) p(h_i). \tag{23}$$

The likelihood measures image agreement; the prior expresses expected orientations, shifts or class frequencies. The normalization sums or integrates over all possible states, including rotations in $SO(3)$.

6.2 From Gaussian noise to weighted least squares

If the vectorized image is $\mathbf{y}_i \in \mathbb{R}^{D^2}$ and predicted image $\boldsymbol{\mu}_i(\theta, h_i)$, assume $\epsilon_i \sim \mathcal{N}(0, \sigma^2 I)$. Then

$$p(\mathbf{y}_i \mid \theta, h_i) = (2\pi\sigma^2)^{-D^2/2} \exp\left(-\frac{\|\mathbf{y}_i - \boldsymbol{\mu}_i\|^2}{2\sigma^2}\right). \tag{24}$$

Taking negative logarithms removes the exponential and yields (up to additive constants) the least-squares objective $\|\mathbf{y}_i - \boldsymbol{\mu}_i\|^2/(2\sigma^2)$. Therefore, least squares is not just a heuristic: it is maximum likelihood under a specified Gaussian-noise model.

6.3 2D classification as a finite mixture

Let $A_1, \ldots, A_K$ be K class-average templates, $\pi_k \geq 0$ their prior probabilities with $\sum_k \pi_k = 1$, and $T_{\mathbf{t}} R_\phi^{(2D)} A_k$ a template rotated in-plane by ϕ and shifted by $\mathbf{t}$. A simplified class likelihood, marginalized over alignment, is

$$p(I_i \mid c_i = k) = \int p(I_i \mid k, \phi, \mathbf{t}) p(\phi, \mathbf{t}) \, \mathrm{d}\phi \, \mathrm{d}\mathbf{t}. \tag{25}$$

Bayes' rule gives the *responsibility* of class k for particle i:

$$\boxed{\gamma_{ik} = p(c_i = k \mid I_i) = \frac{\pi_k p(I_i \mid c_i = k)}{\sum_{\ell=1}^{K} \pi_\ell p(I_i \mid c_i = \ell)}.} \tag{26}$$

For already aligned particles, an idealized class average is $A_k = \left(\sum_i \gamma_{ik} I_i\right) / \left(\sum_i \gamma_{ik}\right)$. Real implementations adjust the alignment, noise and CTF.

Worked example

There are two candidate 2D classes with priors $(\pi_1, \pi_2) = (0.6, 0.4)$. Suppose the image likelihoods at the current alignments are $p(I_i \mid 1) = 0.1$ and $p(I_i \mid 2) = 0.3$ (densities in common arbitrary units). Then

$$\gamma_{i1} = \frac{0.6(0.1)}{0.6(0.1) + 0.4(0.3)} = \frac{0.06}{0.18} = \frac{1}{3}, \quad \gamma_{i2} = \frac{2}{3}.$$

Although class 1 was initially more common, the observed image favors class 2. Hard assignment discards the 1/3 probability; soft assignment retains it.

6.4 Why averaging works: a variance proof

For n correctly aligned images $I_j = s + \epsilon_j$ of the same signal s, suppose $\mathbb{E}[\epsilon_j] = 0$ and the errors are independent with variance σ^2. The average $\overline{I} = s + n^{-1} \sum_j \epsilon_j$ satisfies

$$\operatorname{Var}(\overline{I}) = \operatorname{Var}\left(\frac{1}{n} \sum_j \epsilon_j\right) = \frac{1}{n^2} \sum_j \sigma^2 = \frac{\sigma^2}{n}. \tag{27}$$

Standard deviation decreases as $1/\sqrt{n}$, while the signal remains unchanged. This assumes identically aligned signal and independent noise; wrong alignment or mixtures of conformations can instead blur genuine features.

Worked example

At a single pixel, suppose true signal is 5 and measurement-noise standard deviation is 20. For 400 genuinely equivalent, correctly aligned particles, the average noise standard deviation is $20/\sqrt{400} = 1$. The mean signal stays 5. The amplitude SNR rises from $5/20 = 0.25$ to $5/1 = 5$—a twenty-fold improvement.

6.5 Why 2D classes are not direct conformational states

A rigid asymmetric molecule imaged from the top and from the side naturally forms different 2D averages despite being the *same* 3D conformation. Conversely, distinct conformations can produce deceptively similar 2D views. Classification mixes effects of in-plane alignment, 3D viewing direction, composition, damage, and internal motion. A claim about biologically distinct conformations requires 3D evidence and independent checks.

7 Three-dimensional reconstruction as a linear inverse problem

7.1 Set up the projection matrices

Temporarily assume the pose and CTF of each particle are known. Vectorize the 3D map into $\mathbf{v} \in \mathbb{R}^L$ with $L = D_v^3$, and each observed image into $\mathbf{y}_i \in \mathbb{R}^P$ with $P = D^2$. A linear operator $A_i : \mathbb{R}^L \to \mathbb{R}^P$ performs projection, CTF filtering and translation:

$$\boxed{\mathbf{y}_i = A_i \mathbf{v} + \epsilon_i.} \tag{28}$$

The operator changes with the particle's pose and microscope parameters. The same unknown volume $\mathbf{v}$ must explain all particle images.

7.2 Derive the normal equations

Assuming Gaussian noise, minimize

$$L(\mathbf{v}) = \sum_{i=1}^{N} \|\mathbf{y}_i - A_i\mathbf{v}\|_2^2.$$

Differentiate $\|\mathbf{y} - A\mathbf{v}\|^2 = (\mathbf{y} - A\mathbf{v})^\mathsf{T}(\mathbf{y} - A\mathbf{v})$ with respect to $\mathbf{v}$:

$$\nabla_\mathbf{v} L = -2\sum_i A_i^\mathsf{T}(\mathbf{y}_i - A_i\mathbf{v}).$$

Set the gradient to zero, expand, and rearrange:

$$\boxed{\left(\sum_i A_i^\mathsf{T} A_i\right)\widehat{\mathbf{v}} = \sum_i A_i^\mathsf{T}\mathbf{y}_i.} \tag{29}$$

The right side is a sum of backprojected observations. The left side tells us how often and how strongly volume directions have been sampled. If the matrix is invertible, $\widehat{\mathbf{v}} = (\sum_i A_i^\mathsf{T} A_i)^{-1}\sum_i A_i^\mathsf{T}\mathbf{y}_i$. Actual software avoids explicitly constructing this enormous inverse, using Fourier-domain methods and iterative solvers.

Worked example

Consider a two-unknown volume $\mathbf{v} = (v_1, v_2)^\mathsf{T}$ with measurements $y_1 = v_1 + v_2 = 8$ and $y_2 = v_1 - v_2 = 2$. The combined matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 8 \\ 2 \end{pmatrix}.$$

Compute $A^\mathsf{T}A = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ and $A^\mathsf{T}\mathbf{y} = (10, 6)^\mathsf{T}$. The normal equations give $(v_1, v_2) = (5, 3)$. If only the first measurement existed, A would have rank 1 and infinitely many solutions satisfying $v_1 + v_2 = 8$. This is the simplest algebraic version of the missing-view problem.

7.3 Regularization and maximum a posteriori inference

With additive penalty $\lambda\|L\mathbf{v}\|^2$ (where L can be an identity or a smoothness operator),

$$\widehat{\mathbf{v}} = \arg\min_\mathbf{v} \left[\sum_i \|\mathbf{y}_i - A_i\mathbf{v}\|^2 + \lambda\|L\mathbf{v}\|^2\right], \tag{30}$$

so

$$\left(\sum_i A_i^\mathsf{T} A_i + \lambda L^\mathsf{T} L\right)\widehat{\mathbf{v}} = \sum_i A_i^\mathsf{T}\mathbf{y}_i. \tag{31}$$

Under Gaussian noise and a Gaussian prior $p(\mathbf{v}) \propto \exp[-\lambda\|L\mathbf{v}\|^2/(2\sigma^2)]$, the objective is a negative log posterior. This is *maximum a posteriori* (MAP) estimation. Excessive regularization creates a smooth but biased map; insufficient regularization allows noise to dominate underconstrained frequencies.

7.4 Pose uncertainty: the nonlinear coupled problem

In reality, A_i depends on unknown orientation R_i and translation $\mathbf{t}_i$. The joint problem becomes

$$\min_{V,\{R_i,\mathbf{t}_i\}} \sum_i \|I_i - H_i T_{\mathbf{t}_i} P_{R_i} V\|^2 + \mathcal{R}(V). \tag{32}$$

Rotations live on $SO(3)$, not a linear vector space. The objective is highly nonconvex and can have local optima. *Ab-initio* reconstruction tries to discover a usable volume and poses; subsequent refinement improves them. Their success relies on sufficient shared signal and safeguards against reference bias.

8 Expectation-maximization and Bayesian pose marginalization

8.1 The latent pose posterior

Define $h_i = (R_i, \mathbf{t}_i)$ and parameters θ containing the current volume. The pose posterior is

$$\boxed{p(h_i \mid I_i, \theta) = \frac{p(I_i \mid h_i, \theta)p(h_i)}{\int p(I_i \mid h, \theta)p(h)\,\mathrm{d}h}.} \tag{33}$$

The integral includes orientations and translations, using an appropriate measure over $SO(3)\times\mathbb{R}^2$. The denominator is the marginal image likelihood (also called evidence). A low-SNR particle may have a broad posterior instead of one sharply defined pose.

8.2 EM: an explicit two-step derivation

Introduce an arbitrary normalized variational density $q_i(h)$ and write

$$\log p(I_i \mid \theta) = \log \int q_i(h) \frac{p(I_i, h \mid \theta)}{q_i(h)}\,\mathrm{d}h \tag{34}$$

$$\geq \int q_i(h) \log \frac{p(I_i, h \mid \theta)}{q_i(h)}\,\mathrm{d}h, \tag{35}$$

where the inequality is Jensen's inequality. For fixed θ, the bound is tight when $q_i(h)$ equals the true posterior $p(h \mid I_i, \theta)$. This motivates alternating steps:

$$\begin{aligned} &\text{E-step:} \quad q_i^{(t)}(h) = p(h \mid I_i, \theta^{(t)}), \\ &\text{M-step:} \quad \theta^{(t+1)} = \arg\max_{\theta} \sum_i \int q_i^{(t)}(h) \log p(I_i, h \mid \theta)\,\mathrm{d}h. \end{aligned} \tag{36}$$

Intuitively, E evaluates which poses are plausible given the current map; M updates the map using those pose probabilities. Hard alignment is the limiting approximation in which the posterior is replaced by a delta mass at its most likely pose.

Worked example

For two candidate orientations, suppose likelihoods are $L_1 = 0.18$ and $L_2 = 0.02$ and priors are equal. Then posterior probabilities are $0.18/(0.18 + 0.02) = 0.9$ and 0.1. If the current map changes such that the likelihoods become 0.06 and 0.04, posteriors become 0.6 and 0.4. Assigning orientation 1 with certainty hides the instability visible in the second posterior.

8.3 A gauge ambiguity: absolute orientation cannot be recovered

Suppose $S \in SO(3)$ rotates the entire reference volume and all particle poses consistently. Then the predicted images are unchanged. This global rotation is a *gauge freedom*: the absolute coordinate axes of the reconstructed map have no physical meaning without an external reference. A related handedness/mirror ambiguity arises under projection geometry and must be resolved using additional structural information or other constraints. Neither should be mistaken for an apparent physical deformation of PP2A.

9 Resolution, validation, and whether the map is trustworthy

9.1 Gold-standard half-maps

Split particles into two independent halves and reconstruct V_A and V_B while minimizing information leakage. Transform to Fourier coefficients $\widehat{V}_A(\mathbf{K}), \widehat{V}_B(\mathbf{K})$. On a shell S_k of frequencies near magnitude k, define

$$\boxed{\mathrm{FSC}(k) = \frac{\sum_{\mathbf{K}\in S_k} \widehat{V}_A(\mathbf{K})\widehat{V}_B(\mathbf{K})^*}{\sqrt{\left(\sum_{S_k} |\widehat{V}_A|^2\right)\left(\sum_{S_k} |\widehat{V}_B|^2\right)}}.} \tag{37}$$

* means complex conjugate. For real-space maps, conjugate frequency symmetry makes the shell expression real up to numerical error; practical implementations may use its real part. The FSC compares independent half reconstructions in bands of spatial frequency. A conventional global resolution uses the frequency k_* where FSC crosses 0.143 and reports $d_* = 1/k_*$.

9.2 Deriving the FSC–SNR relation under simple assumptions

At a given Fourier shell, model independent half maps as $A = S + N_1$ and $B = S + N_2$, where S is common signal, noises are zero-mean independent of each other and of S, and expected powers are $P_S = \mathbb{E}|S|^2$, $P_N = \mathbb{E}|N_1|^2 = \mathbb{E}|N_2|^2$. Then

$$\mathbb{E}[AB^*] = P_S, \qquad \mathbb{E}|A|^2 = \mathbb{E}|B|^2 = P_S + P_N.$$

Thus an idealized population FSC is

$$\mathrm{FSC} = \frac{P_S}{P_S + P_N} = \frac{\mathrm{SNR}_{\mathrm{half}}}{1 + \mathrm{SNR}_{\mathrm{half}}}, \quad \mathrm{SNR}_{\mathrm{half}} = \frac{\mathrm{FSC}}{1 - \mathrm{FSC}}. \tag{38}$$

This is a relation of expected powers under the stated independence assumptions, not an exact identity for any finite noisy map pair.

Worked example

For FSC = 0.143, the half-map power SNR estimate is $0.143/(1 - 0.143) = 0.1669 \approx 1/6$. Two halves combined (with independent equal-variance noise) double the power SNR, giving a full-data SNR near 1/3 at that frequency. This explains the statistical rationale behind the conventional 0.143 criterion; it does not prove every feature at the reported resolution is correct.

9.3 Directional, local and model-to-map validation

FSC averages information over all Fourier directions, so it can hide severe anisotropy caused by preferred orientation. Directional FSC restricts frequencies to angular cones; local-resolution analysis tests spatially restricted map portions. Atomic validation additionally considers stereochemical geometry, local fit, independent half-map consistency, and potential overfitting. A high global FSC number does not establish that every HEAT repeat or attachment region is resolved at the same quality.

Important for isolated PP2A A-subunit. Its modest mass, extended repeat architecture, internal flexibility and potentially flexible DNA make joint orientation/conformation identifiability difficult. A change in pose distribution or flexible-appendage visibility between conditions can resemble a change in protein curvature. Compare protein-centric and whole-particle alignment as a diagnostic; use directional diagnostics as well as global FSC.

10 Why a single map can hide meaningful conformational variation

10.1 Discrete versus continuous heterogeneity

A homogeneous model uses $I_i \approx A_i V_0 + \epsilon_i$. A discrete mixture replaces V_0 with V_{c_i}, where class c_i takes one of K values. A continuous model instead writes $V(\mathbf{x}; z_i)$, where z_i is a continuous conformational vector. Both types of

variation can coexist. For example, a regulatory subunit can be present or absent (composition), while the PP2A scaffold bends continuously (conformation). Treating them as one kind of variability can produce intermediate "half-occupied" density maps that are not real molecular species.

10.2 A mixture model for three-dimensional classes

Assume K volumes V_k and class proportions π_k. Given pose $h_i = (R_i, \mathbf{t}_i)$,

$$p(I_i) = \sum_{k=1}^{K} \pi_k \int p(I_i \mid V_k, h) p(h \mid k)\, \mathrm{d}h. \tag{39}$$

For fixed poses, the posterior class probability is

$$\gamma_{ik} = \frac{\pi_k p(I_i \mid V_k, h_i)}{\sum_{\ell=1}^{K} \pi_\ell p(I_i \mid V_\ell, h_i)}. \tag{40}$$

These soft assignments permit reconstruction of separate volumes from weighted particle contributions. However, if poses are biased by a consensus map, classification performed without pose re-estimation inherits that bias. Some algorithms jointly update poses and classes; others fix prior alignments.

Worked example

Imagine 1,000 actual particles, 700 in conformation A and 300 in B, with identical picking probability. If assignments were perfect, the estimated mixture fractions would be $(0.7, 0.3)$. But if state B is picked only half as efficiently, the observed sample fractions become $700/(700+150) = 0.8235$ and $150/(850) = 0.1765$. Classification may perfectly separate the accepted particles and still produce *biased biological populations*.

11 Covariance, PCA, and CryoSPARC 3D variability analysis

11.1 What is a covariance matrix in volume space?

Imagine a hypothetical known 3D potential volume $\mathbf{v}_i \in \mathbb{R}^L$ for each particle, $L = D_v^3$. Define

$$\overline{\mathbf{v}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{v}_i, \quad C = \frac{1}{N-1}\sum_{i=1}^{N} (\mathbf{v}_i - \overline{\mathbf{v}})(\mathbf{v}_i - \overline{\mathbf{v}})^{\mathsf{T}}. \tag{41}$$

$C \in \mathbb{R}^{L \times L}$; its entry C_{ab} is the covariance between potential values at voxels a and b. It is unrealistic to explicitly construct this giant matrix for a large box. Low-rank methods instead estimate a small number of leading modes from the 2D images.

11.2 PCA from a constrained optimization proof

The direction $\mathbf{u}$ of maximum projected variance solves

$$\max_{\mathbf{u}} \ \mathbf{u}^{\mathsf{T}} C \mathbf{u} \quad \text{subject to} \quad \mathbf{u}^{\mathsf{T}}\mathbf{u} = 1. \tag{42}$$

Introduce Lagrange multiplier λ:

$$\mathcal{J}(\mathbf{u}, \lambda) = \mathbf{u}^{\mathsf{T}} C \mathbf{u} - \lambda(\mathbf{u}^{\mathsf{T}}\mathbf{u} - 1).$$

Differentiating with respect to $\mathbf{u}$ gives $2C\mathbf{u} - 2\lambda\mathbf{u} = 0$, or

$$\boxed{C\mathbf{u} = \lambda \mathbf{u}.} \tag{43}$$

Thus the principal components are eigenvectors of the covariance. Subsequent components maximize remaining variance subject to orthogonality to earlier components. This derivation reveals precisely why PCA modes optimize *variance*, not physical energy, force sensitivity, or causal relevance.

11.3 The linear low-rank volume model

With orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_K$,

$$\mathbf{v}_i \approx \overline{\mathbf{v}} + \sum_{j=1}^{K} a_{ij}\mathbf{u}_j, \qquad a_{ij} = \mathbf{u}_j^{\mathsf{T}}(\mathbf{v}_i - \overline{\mathbf{v}}). \tag{44}$$

The coefficients a_{ij} are PCA scores. Each $\mathbf{u}_j$ can be reshaped into a signed 3D variability map U_j. The 2D cryo-EM model is then

$$I_i \approx A_i\left(V_0 + \sum_{j=1}^{K} a_{ij}U_j\right) + \epsilon_i. \tag{45}$$

The measurement operator A_i projects and CTF-filters the volume. CryoSPARC 3DVA operates conceptually in this family of linear variability models using previously estimated particle alignments. The alignment assumption matters: sufficiently dissimilar conformations can cause the consensus alignment to fail. See the official CryoSPARC guide, CryoSPARC 3DVA guide (online).

11.4 Fully worked numerical PCA example

Worked example

Suppose an imaginary three-voxel object fluctuates in only two dimensions, with centered volume deviations

$$\delta_1 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \quad \delta_2 = \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix}, \quad \delta_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \delta_4 = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}.$$

These deviations sum to zero, so the mean deviation is zero. With the sample normalization $1/(N-1) = 1/3$, the covariance is

$$C = \frac{1}{3}\sum_{i=1}^{4} \delta_i\delta_i^{\mathsf{T}} = \begin{pmatrix} 8/3 & 0 & 0 \\ 0 & 2/3 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Eigenvectors are the coordinate axes $\mathbf{u}_1 = (1, 0, 0)$ and $\mathbf{u}_2 = (0, 1, 0)$ with eigenvalues $8/3$ and $2/3$. Thus the first mode explains four times as much sample variance as the second. It does *not* imply that the first mode is four times as important for biological activity.

11.5 Why linear density variation approximates small motion

If a one-dimensional structure translates by displacement q, the true density is $\rho_q(x) = \rho_0(x - q)$. Taylor's theorem gives

$$\rho_0(x - q) = \rho_0(x) - q\rho_0'(x) + \frac{q^2}{2}\rho_0''(x) + O(q^3). \tag{46}$$

A single linear mode $U(x) = -\rho_0'(x)$ captures the first-order term. When $|q|$ is small relative to the spatial scale of density features, the approximation is good. For large displacements, higher-order terms become important; a fixed linear mode can generate negative density and misleading apparent feature splitting. For large PP2A bending, a one-dimensional linear movie is a visualization, not proof of an actual atomic pathway.

11.6 Why an apparently smooth mode can fold back

Consider a true two-dimensional curved conformational trajectory $\mathbf{v}(q) = (\cos q, \sin q)^{\mathsf{T}}$. Project it onto the first coordinate axis to get $a(q) = \cos q$. Then $a(\pi/3) = a(-\pi/3) = 1/2$, even though the corresponding physical conformations are different. The inferred coordinate is many-to-one. Moreover $a'(q) = -\sin q$ changes sign. A sequence of increasing physical q therefore first increases then decreases the inferred a.

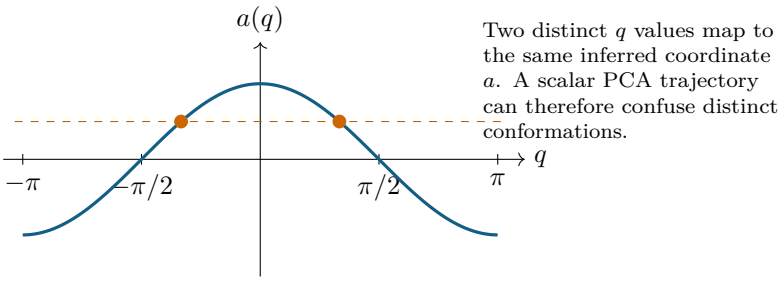


Figure 4: Analytic illustration of coordinate folding. This is not a measured cryo-EM result.

Worked example

If physical q takes values $0, \pi/3, 2\pi/3, \pi$, then the inferred coordinate $a = \cos q$ is $(1, 1/2, -1/2, -1)$, monotone. But if the trajectory spans $-\pi$ to π, a goes $(-1, 1, -1)$; neither the sign nor ordering of q can be reconstructed globally from a alone. Whether folding occurs depends on the sampled domain, geometry and projection, not merely on the presence of nonlinearity.

12 RECOVAR and statistically estimated conformational covariance

RECOVAR is a regularized covariance/PCA approach that reconstructs heterogeneity and employs adaptive kernel regression for state-specific volumes. Its motivation is that a latent space with meaningful geometric structure can support more controlled population analysis than arbitrary nonlinear embeddings. The 2025 PNAS paper is Gilles and Singer, *Cryo-EM heterogeneity analysis using regularized covariance estimation and kernel regression*, DOI: 10.1073/pnas.2419140122.

12.1 Second-order moments in the image model

Use the centered linear imaging model

$$\mathbf{y}_i = A_i \mathbf{v}_i + \epsilon_i, \qquad \mathbb{E}[\mathbf{v}_i] = \overline{\mathbf{v}}, \quad \mathrm{Cov}(\mathbf{v}_i) = C.$$

Under independence of molecular state and additive zero-mean noise, conditioned on known A_i,

$$\mathbb{E}[\mathbf{y}_i \mid A_i] = A_i \overline{\mathbf{v}}, \qquad \mathrm{Cov}(\mathbf{y}_i \mid A_i) = A_i C A_i^{\mathsf{T}} + \Sigma_{\epsilon,i}. \tag{47}$$

The identity follows from $\mathrm{Cov}(A\mathbf{v}) = A\,\mathrm{Cov}(\mathbf{v})A^{\mathsf{T}}$. The inverse covariance problem tries to infer C from the family of *projected* covariance observations. Regularization is essential because many covariance degrees of freedom are weakly constrained.

12.2 Scores with noise-aware projections

Once a low-rank basis $U \in \mathbb{R}^{L\times K}$ is estimated, write $\mathbf{v}_i \approx \overline{\mathbf{v}} + U\mathbf{a}_i$. An idealized weighted ridge estimate is

$$\widehat{\mathbf{a}}_i = \arg\min_{\mathbf{a}} \left[\|\Sigma_i^{-1/2}(\mathbf{y}_i - A_i\overline{\mathbf{v}} - A_i U\mathbf{a})\|^2 + \mathbf{a}^{\mathsf{T}}\Lambda^{-1}\mathbf{a} \right]. \tag{48}$$

Setting the gradient to zero gives

$$\widehat{\mathbf{a}}_i = \left(U^{\mathsf{T}}A_i^{\mathsf{T}}\Sigma_i^{-1}A_iU + \Lambda^{-1}\right)^{-1} U^{\mathsf{T}}A_i^{\mathsf{T}}\Sigma_i^{-1}(\mathbf{y}_i - A_i\overline{\mathbf{v}}). \tag{49}$$

Λ is a covariance/regularization matrix for the score prior. This is a generic noise-aware low-rank inference formula, not a claim that it reproduces every implementation detail of RECOVAR. It shows how uncertain viewing directions and weak signal can shrink estimated scores toward the prior.

12.3 Kernel regression for state-conditioned volumes

Given latent coordinates $\mathbf{a}_i$ and a target coordinate $\mathbf{a}_0$, define kernel weights

$$w_i(\mathbf{a}_0) = K_h(\mathbf{a}_i - \mathbf{a}_0).$$

A weighted reconstruction is then

$$\widehat{\mathbf{v}}(\mathbf{a}_0) = \arg\min_{\mathbf{v}} \left[\sum_i w_i(\mathbf{a}_0) \|\mathbf{y}_i - A_i\mathbf{v}\|^2_{\Sigma_i^{-1}} + \mathcal{R}(\mathbf{v}) \right]. \tag{50}$$

K_h is a kernel of bandwidth h. Nearby latent states contribute more strongly; smaller bandwidth gives greater structural specificity but fewer effective particles and potentially higher variance. Larger bandwidth reduces noise while averaging together distinct conformations.

Worked example

Suppose a target state $a_0 = 0$ and particle coordinates $(0, 1, 2)$, with Gaussian weights $w_i = \exp[-(a_i - a_0)^2/(2h^2)]$ and $h = 1$. The weights are $(1, 0.607, 0.135)$. Normalized weights are approximately $(0.574, 0.348, 0.078)$. The nearest state dominates, but other particles still inform the reconstruction. Shrinking h to 0.5 gives much less weight to the distant particles and a noisier but more localized estimate.

13 Nonlinear neural models: cryoDRGN and variational inference

13.1 A nonlinear family of volumes

Instead of $V_0 + \sum_j a_j U_j$, define $V_\theta(\mathbf{x}, z)$, with neural parameters θ and latent $z \in \mathbb{R}^d$. In Fourier-space implementations a decoder produces $f_\theta(\mathbf{K}, z)$, the complex Fourier coefficient of the conformation-z volume at frequency $\mathbf{K}$. Evaluating the decoder on the particle's central plane yields a predicted particle image through Equation (21).

13.2 Encoder and decoder input/output

The encoder receives a preprocessed image I_i and outputs parameters of a latent posterior $q_\phi(z \mid I_i)$, for example mean $\boldsymbol{\mu}_i \in \mathbb{R}^d$ and diagonal standard deviations $\boldsymbol{\sigma}_i \in \mathbb{R}^d$. A latent sample z_i enters the decoder, together with frequency coordinates, and produces predicted Fourier slices. The loss compares these predictions to observed images, using the known CTF and previously estimated poses where the method assumes fixed poses. Original cryoDRGN used consensus-derived poses and a variational encoder–decoder architecture (Zhong et al., *Nature Methods*, 2021, DOI: 10.1038/s41592-020-01049-4).

13.3 Derive the evidence lower bound (ELBO)

For latent prior $p(z)$ and image likelihood $p_\theta(I \mid z)$, the observed likelihood is $p_\theta(I) = \int p_\theta(I \mid z)p(z)\,\mathrm{d}z$. Multiply and divide the integrand by the approximate posterior $q_\phi(z \mid I)$, then apply Jensen's inequality:

$$\log p_\theta(I) = \log \mathbb{E}_{q_\phi}\left[\frac{p_\theta(I \mid z)p(z)}{q_\phi(z \mid I)}\right] \tag{51}$$

$$\geq \mathbb{E}_{q_\phi}[\log p_\theta(I \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid I)\|p(z)). \tag{52}$$

The KL divergence is $D_{\mathrm{KL}}(q\|p) = \int q(z)\log[q(z)/p(z)]\,\mathrm{d}z$. The first term rewards accurate image prediction; the second regularizes the posterior toward the prior. The negative right-hand side is a common VAE training loss. Crucially, the likelihood decoder is constrained by physical projection geometry, not an arbitrary image-to-image transformation.

13.4 A worked scalar Gaussian KL example

For $q(z) = \mathcal{N}(\mu, \sigma^2)$ and prior $p(z) = \mathcal{N}(0, 1)$,

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\left(\mu^2 + \sigma^2 - 1 - \log\sigma^2\right). \tag{53}$$

Worked example

If the encoder predicts $\mu = 1$ and $\sigma = 0.5$, then $\sigma^2 = 0.25$ and

$$D_{\mathrm{KL}} = \frac{1}{2}[1 + 0.25 - 1 - \log(0.25)] = \frac{1}{2}[0.25 + 1.3863] \approx 0.8181.$$

This costs about 0.818 nats of regularization. The optimizer must improve reconstruction enough to justify such a departure from the unit-normal prior. The numerical value is not an experimental energy in k_BT units.

13.5 The reparameterization trick

To backpropagate through Gaussian latent sampling, write

$$z = \mu_\phi(I) + \sigma_\phi(I) \odot \eta, \qquad \eta \sim \mathcal{N}(0, I).$$

Here $\odot$ denotes elementwise multiplication. Randomness is carried by η while μ_ϕ and σ_ϕ remain differentiable network outputs. The technique estimates gradients of the expectation with low variance using standard automatic differentiation.

13.6 Why latent geometry is not automatically physical

The encoder learns whatever numerical coordinates explain images well under its model and regularizer. Any invertible nonlinear reparameterization can describe the same family of structures with changed coordinate density. Latent distance need not equal RMSD, mechanical extension, or energy. Furthermore, an image may leave parts of the 3D state ambiguous. An apparently precise point estimate z_i can hide broad posterior uncertainty.

14 3DFlex: learning deformations rather than arbitrary densities

14.1 Canonical map, latent state and displacement field

The 3DFlex family assumes that flexible conformations are generated by deforming a shared canonical map V_0. A flow generator produces a displacement or velocity field $\mathbf{u}_\theta(\mathbf{x}, z) \in \mathbb{R}^3$ from the latent z, and a deformation/convection operator yields $V_z = \mathcal{D}(V_0, \mathbf{u}_\theta(z))$. The observation model is

$$I_i \approx H_i T_{\mathbf{t}_i} P_{R_i} \mathcal{D}(V_0, \mathbf{u}_\theta(z_i)) + \epsilon_i. \tag{54}$$

The exact numerical convection convention matters; the equation here is schematic. The method is described by Punjani and Fleet, *Nature Methods* (2023), DOI: 10.1038/s41592-023-01853-8. Its canonical-deformation prior can be effective for continuous motion but less appropriate for density addition/removal, for example severe compositional heterogeneity.

14.2 Geometric warp versus density transport

For a deformation $\mathbf{y} = \Phi_z(\mathbf{x})$, a geometric pullback of scalar values can be written $V_z(\mathbf{y}) = V_0(\Phi_z^{-1}(\mathbf{y}))$. If V is interpreted as a conserved mass density, the correct transport instead also involves a Jacobian determinant:

$$V_z(\mathbf{y}) = \frac{V_0(\Phi_z^{-1}(\mathbf{y}))}{|\det D\Phi_z(\Phi_z^{-1}(\mathbf{y}))|}.$$

Which representation is appropriate depends on the physical interpretation and implemented convection operator. The distinction is worth knowing when comparing generic deformable-image networks to physics-based transport models. One must also regularize unrealistic local strains or folding deformations.

14.3 A tiny worked deformation example

Let a 1D density be $V_0(x) = \exp[-x^2/(2\sigma^2)]$. A rigid translation $\Phi_q(x) = x+q$ yields $V_q(x) = \exp[-(x-q)^2/(2\sigma^2)]$. A linear 3DVA approximation would instead use $V_0(x) - qV_0'(x)$. With $\sigma = 1$, $x = q = 1$, the true shifted value is 1; the linear approximation is $e^{-1/2} + e^{-1/2} = 1.2131$, a 21.3% error at that coordinate. This illustrates the failure of first-order density modes at finite displacement (not a quantified error for 3DFlex).

15 Physical structure metrics and atomic-model comparisons

15.1 Volume-derived observables versus atom-derived observables

A 3D map permits measurements of shapes, densities, and fitted domain positions. Atomic-model metrics additionally require a reliable fitted set of coordinates $\mathbf{r}_1, \ldots, \mathbf{r}_M$. Two maps reconstructed at different local resolutions should not be compared as though their fine-scale details were equally trustworthy. Before computing apparent displacement, register structures into a common coordinate system and decide whether alignment should use the whole protein or a mechanically stable core. For a bent scaffold, full-protein alignment can partly hide the motion of interest.

15.2 RMSD and the rigid-alignment problem

Given corresponding M atomic coordinates $\mathbf{a}_j$ and $\mathbf{b}_j$ from two conformations, the naive root-mean-square deviation is

$$\mathrm{RMSD}_{\mathrm{raw}} = \sqrt{\frac{1}{M}\sum_{j=1}^{M} \|\mathbf{a}_j - \mathbf{b}_j\|^2}. \tag{55}$$

This is meaningless as an internal deformation statistic if one structure was arbitrarily rotated or translated. Instead define the aligned RMSD

$$\mathrm{RMSD}_{\mathrm{fit}} = \min_{R\in SO(3),\, \mathbf{t}\in\mathbb{R}^3} \sqrt{\frac{1}{M}\sum_{j} \|R\mathbf{a}_j + \mathbf{t} - \mathbf{b}_j\|^2}. \tag{56}$$

The minimization removes global rigid-body motion while preserving internal differences.

15.3 Deriving Kabsch alignment via the SVD

Let centroids be $\bar{\mathbf{a}} = M^{-1}\sum_j \mathbf{a}_j$ and $\bar{\mathbf{b}} = M^{-1}\sum_j \mathbf{b}_j$. For fixed R, differentiating the squared residual with respect to $\mathbf{t}$ gives optimum $\mathbf{t}^* = \bar{\mathbf{b}} - R\bar{\mathbf{a}}$. Therefore center the coordinates $\mathbf{x}_j = \mathbf{a}_j - \bar{\mathbf{a}}$ and $\mathbf{y}_j = \mathbf{b}_j - \bar{\mathbf{b}}$. Expand:

$$\sum_j \|R\mathbf{x}_j - \mathbf{y}_j\|^2 = \sum_j \|\mathbf{x}_j\|^2 + \sum_j \|\mathbf{y}_j\|^2 - 2\sum_j \mathbf{y}_j^{\mathsf{T}} R\mathbf{x}_j. \tag{57}$$

The first two terms do not depend on R. Define $H = \sum_j \mathbf{x}_j\mathbf{y}_j^{\mathsf{T}}$; the last sum is $\mathrm{tr}(RH)$. The problem is therefore maximizing $\mathrm{tr}(RH)$ over $SO(3)$. Compute an SVD $H = U\Sigma V^{\mathsf{T}}$. One maximizing proper rotation is

$$\boxed{R^* = V\,\mathrm{diag}(1, 1, \det(VU^{\mathsf{T}}))U^{\mathsf{T}}.} \tag{58}$$

The determinant adjustment disallows a reflection. This proves why RMSD comparisons must optimize a global pose, and why the chosen aligned atom subset matters.

Worked example

Take two points $\mathbf{a}_1 = (0,0,0)$, $\mathbf{a}_2 = (2,0,0)$ and a second configuration $\mathbf{b}_1 = (1,1,0)$, $\mathbf{b}_2 = (1,3,0)$. Without alignment, RMSD is nonzero. But the second pair is exactly the first pair rotated 90 degrees in the xy plane and translated by $(1,1,0)$. The fitted RMSD is exactly zero: there was no internal deformation at all.

15.4 Distances and inter-repeat angles

For two attachment positions $\mathbf{r}_A(q)$ and $\mathbf{r}_B(q)$,

$$d(q) = \|\mathbf{r}_B(q) - \mathbf{r}_A(q)\|. \tag{59}$$

A distance is invariant to global rotation and translation. For adjacent HEAT repeats with fitted rotations R_j, R_{j+1} relative to a common reference, the relative rotation is $Q_j = R_j^{\mathsf{T}} R_{j+1}$. Its rotation angle is

$$\theta_j = \cos^{-1}\left(\frac{\operatorname{tr}(Q_j) - 1}{2}\right), \quad \theta_j \in [0, \pi]. \tag{60}$$

Because $\operatorname{tr}(Q) = 1 + 2\cos\theta$ for a 3D proper rotation, the formula follows directly. In code the inverse-cosine argument should be clipped to $[-1, 1]$ against floating-point error. Torsion requires an additional choice of local axes, and cannot be inferred uniquely from a single scalar rotation angle.

16 Statistical mechanics: what population densities mean

16.1 Physical coordinates, densities, and histograms

Let q be a physically defined coordinate, such as PP2A attachment-site separation in nm. Its probability *density* $p(q)$ satisfies $\int p(q)\,\mathrm{d}q = 1$. The probability of interval $[a, b]$ is $\int_a^b p(q)\,\mathrm{d}q$. Density has reciprocal units of q. A kernel-density estimator is

$$\widehat{p}(q) = \frac{1}{Nh}\sum_{i=1}^{N} K\left(\frac{q - q_i}{h}\right), \tag{61}$$

where K is a normalized kernel and h is a bandwidth with the same units as q. Assigning each particle a single q_i overlooks inferential uncertainty; integrating particle posteriors $p(q \mid I_i)$ is preferable when these are broad.

16.2 Deriving free energy from equilibrium statistics

At thermal equilibrium, a potential of mean force $G(q)$ satisfies

$$p(q) = \frac{1}{Z}\exp[-\beta G(q)], \qquad \beta = \frac{1}{k_B T}, \quad Z = \int \exp[-\beta G(q)]\,\mathrm{d}q. \tag{62}$$

Take logarithms: $\log p(q) = -\beta G(q) - \log Z$. Rearranging yields

$$\boxed{G(q) = -k_B T \log p(q) + C,} \tag{63}$$

with arbitrary additive constant $C = -k_B T \log Z$. k_B is Boltzmann's constant; T the absolute temperature; β inverse thermal energy. This is valid only if the inferred $p(q)$ represents the correct equilibrium distribution under a known coordinate measure and appropriate unbiased sampling.

Worked example

In a two-state equilibrium, suppose probabilities are $P_A = 0.8$, $P_B = 0.2$. The state free-energy difference $\Delta G = G_B - G_A$ is

$$\Delta G = -k_B T \log(P_B/P_A) = -k_B T \log(0.25) = 1.3863\, k_B T.$$

State B is higher in free energy by 1.3863 thermal units, despite existing in 20% of the ensemble. This assumes both states are observed proportionally to their equilibrium populations.

16.3 The Jacobian problem for latent variables

If $q = f(z)$ is a differentiable bijection, probability conservation gives $p_q(q)\,\mathrm{d}q = p_z(z)\,\mathrm{d}z$, hence

$$p_q(q) = p_z(z)\left|\frac{\mathrm{d}z}{\mathrm{d}q}\right|. \tag{64}$$

In multiple dimensions, the derivative becomes the determinant of the inverse Jacobian. If the mapping is not one-to-one, sum the contributions of all inverse branches. This *measure factor* means that the density of a neural latent variable is not an invariant physical density. A good reconstruction model can still have a latent coordinate unsuitable for $-k_BT\log p$ interpretation.

Worked example

Suppose z is uniform on $[0,1]$ and define $q = z^2$. Then $p_z(z) = 1$, $z = \sqrt{q}$, and $|dz/dq| = 1/(2\sqrt{q})$. Therefore $p_q(q) = 1/(2\sqrt{q})$ for $0 < q < 1$. The q-density is strongly concentrated near zero even though z is uniform. The apparent profile $-\log p_q(q)$ is not flat. It would be wrong to attribute this difference to new physical forces: it arises from reparameterization alone.

16.4 Selection bias and pose-dependent detection

Let $p_{\mathrm{eq}}(q)$ be equilibrium abundance and $w(q)$ be the probability that a particle in state q survives specimen preparation, orientation constraints, picking, alignment and classification. The observed distribution is

$$\boxed{p_{\mathrm{obs}}(q) = \frac{w(q)p_{\mathrm{eq}}(q)}{\int w(q')p_{\mathrm{eq}}(q')\,\mathrm{d}q'}.} \tag{65}$$

Unless $w(q)$ is constant or estimable, p_{obs} does not uniquely determine p_{eq}. A learned network can reconstruct the accepted dataset perfectly and still misrepresent equilibrium populations. This is an identifiability limitation, not a training bug.

Worked example

Suppose true equilibrium populations are $P_A = P_B = 0.5$. Picking accepts A at probability 0.9 and B at 0.3. Accepted proportions are $0.9(0.5)/[0.9(0.5) + 0.3(0.5)] = 0.75$ for A and 0.25 for B. A naive Boltzmann inference reports $G_B - G_A = -k_BT\log(1/3) = 1.099\,k_BT$, even though the actual two-state energy difference was zero. The entire apparent energy difference arose from selection bias.

17 Mechanical perturbations and the DNA-spring potential

17.1 Generalized force and molecular conformations

For a continuous physical coordinate q, let $G_0(q)$ be the relaxed protein potential of mean force. When an externally controlled perturbation c adds energy $U_c(q)$, the total energy is

$$G_c(q) = G_0(q) + U_c(q), \qquad p(q \mid c) = \frac{e^{-\beta[G_0(q)+U_c(q)]}}{Z_c}, \tag{66}$$

where $Z_c = \int e^{-\beta[G_0(q)+U_c(q)]}\,\mathrm{d}q$. The effect is a redistribution of statistical weights; it does not imply that every molecule experiences the same extension or occupies one unique conformation.

17.2 Constant force versus fixed-spring geometry

In an ideal constant-force ensemble with external force F conjugate to extension $x(q)$, the mechanical work term is $U_F(q) = -Fx(q)$. The force *stabilizes* more-extended states by an amount Fx. By contrast, a spring constrained by protein geometry usually produces a coordinate-dependent force and has stored energy, idealized as

$$U_{\mathrm{harm}}(q) = \frac{1}{2}k_s[d(q) - d_0]^2, \qquad F_d = -\frac{\partial U}{\partial d} = -k_s(d - d_0). \tag{67}$$

k_s is stiffness, d_0 preferred separation, and $d(q)$ actual attachment separation. Because DNA behaves as a semiflexible polymer and can bend, twist and kink, the harmonic approximation may be inappropriate for a bent 50- or 80-bp duplex.

17.3 Bending energy of a worm-like chain

For DNA contour length L and arclength $s \in [0, L]$, let $\mathbf{t}(s)$ be the unit tangent to its centerline and ℓ_p its persistence length. In the worm-like-chain bending approximation,

$$E_{\mathrm{bend}} = \frac{k_B T \ell_p}{2} \int_0^L \left\| \frac{\mathrm{d}\mathbf{t}(s)}{\mathrm{d}s} \right\|^2 \mathrm{d}s. \tag{68}$$

The tangent derivative measures local curvature (units of inverse length); $k_B T \ell_p$ is the bending rigidity (energy times length). Actual constrained free energy integrates over permissible DNA conformations and linker states, not merely the minimum bending energy. Kinking and protein–DNA contacts can change the relationship between geometry and transmitted force substantially.

Mechanical calibration is not automatic. An 80-bp spring has a contour length around 27 nm using the conventional dsDNA rise of approximately 0.34 nm per base pair, but this alone does not determine the force it exerts across PP2A attachment sites. Linker compliance, endpoint orientation, kinks, spring geometry and interaction energies remain unknown. These must be modeled or independently calibrated before reporting absolute piconewton forces.

17.4 A fully worked two-state force response

Consider a two-state toy protein with states A and B. Without force, set $G_A(0) = 0$ and $G_B(0) = 2k_B T$. Let the more-extended state B have extension $x_B - x_A = 1.0$ nm. In a constant-force ensemble,

$$G_B(F) - G_A(F) = 2k_B T - F(1.0\,\mathrm{nm}).$$

At $T = 298$ K, $k_B T \approx 4.114$ pN · nm. The population of B is

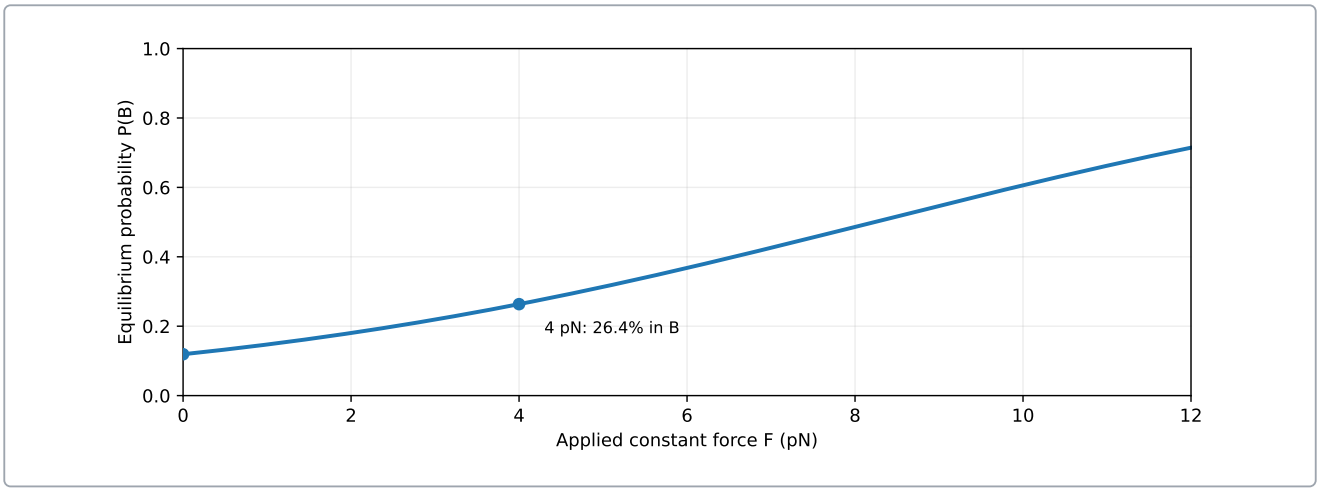
$$P_B(F) = \frac{1}{1 + \exp[\beta \Delta G(F)]}. \tag{69}$$

Worked example

At zero force, $P_B(0) = 1/(1+e^2) \approx 0.1192$. At $F = 4$ pN, mechanical work is $F\Delta x = 4\,\mathrm{pN} \cdot \mathrm{nm} \approx 0.9723\,k_B T$. Thus $\Delta G(4\,\mathrm{pN}) \approx 1.0277\,k_B T$, and

$$P_B(4\,\mathrm{pN}) = \frac{1}{1 + e^{1.0277}} \approx 0.2635.$$

Force shifts B's population from about 12% to 26% without forcing all molecules into B. These are *synthetic illustrative numbers*, not a measurement or force prediction for PP2A DNA chimeras.



17.5 Deriving fluctuation–response relations

In a constant-force equilibrium ensemble $p_F(q) \propto \exp[-\beta(G_0(q) - Fx(q))]$, the mean of observable $A(q)$ is $\langle A\rangle_F = \int A(q)p_F(q)\,\mathrm{d}q$. Differentiate p_F with respect to F while also differentiating its normalization $Z(F)$:

$$\frac{\partial p_F(q)}{\partial F} = \beta p_F(q)[x(q) - \langle x\rangle_F].$$

Then

$$\frac{\partial\langle A\rangle_F}{\partial F} = \int A(q)\frac{\partial p_F(q)}{\partial F}\,\mathrm{d}q \tag{70}$$

$$= \beta[\langle Ax\rangle_F - \langle A\rangle_F\langle x\rangle_F] = \beta\,\mathrm{Cov}_F(A,x). \tag{71}$$

For $A = x$,

$$\boxed{\frac{\partial\langle x\rangle_F}{\partial F} = \beta\,\mathrm{Var}_F(x).} \tag{72}$$

This is an equilibrium fluctuation–response theorem under *constant force*. A finite DNA spring has its own potential $U_c(q)$ and generally requires differentiating with respect to the experiment's actual control parameter, not simply inserting a constant F.

18 SpringEM as a conditioned inverse problem, not just a prettier movie

18.1 Generative model with experimental conditions

Let c_i label the independently prepared condition of particle i (for example an ssDNA chimera, nicked/gapped duplex or intact duplex). The imaging likelihood remains controlled by projection geometry, whereas the conformation prior is conditioned on c_i:

$$\boxed{p_\theta(I_i, z_i, h_i \mid c_i) = p_\theta(I_i \mid z_i, h_i, H_i)\,p_\theta(z_i \mid c_i)\,p(h_i \mid c_i).} \tag{73}$$

h_i is particle pose and H_i its CTF (known or estimated). Including $p(h_i \mid c_i)$ explicitly reminds us that preferred orientations may differ between preparations. The central inference is a condition-specific distribution $p(q \mid c)$ over *physical* coordinates, alongside a common structural image-formation model. A mechanical prior of the form $p(q \mid c) \propto \exp[-\beta(G_0(q) + U_c(q))]$ is meaningful only with a calibrated perturbation potential and equilibrium assumptions.

18.2 What can be identified from two or more conditions?

Suppose $w_c(q)$ denotes condition-specific selection probability. Observed populations satisfy $p_{\mathrm{obs}}(q \mid c) \propto w_c(q)\exp[-\beta(G_0(q)+U_c(q))]$. For two conditions a, b,

$$\log \frac{p_{\mathrm{obs}}(q \mid b)}{p_{\mathrm{obs}}(q \mid a)} = -\beta[U_b(q) - U_a(q)] + \log \frac{w_b(q)}{w_a(q)} + C_{ab}, \tag{74}$$

where C_{ab} is independent of q and collects normalization factors. The unknown intrinsic landscape $G_0(q)$ cancels from the ratio. If selection biases are matched or controlled, the *shape* of the log-population ratio tests the difference in perturbation potentials. If the bias ratio is unknown, an energy-bias explanation is not uniquely identifiable.

Worked example

Suppose equal-selection, equilibrium observations give $P_A/P_B = 4$ in control and $P_A/P_B = 1$ under loading. The loading shifted the state log-odds by $\log[(P_B/P_A)_{\mathrm{load}}/(P_B/P_A)_{\mathrm{ctrl}}] = \log 4 \approx 1.3863$. Therefore the relative B-versus-A mechanical bias changed by $-1.3863\,k_BT$. This says *how strongly populations were reweighted*; it does not by itself identify the force in pN without an extension difference or spring calibration.

18.3 Why PP2A controls matter mechanistically

For the engineered A-subunit with DNA attachment near D6 and N552, comparing wild-type apo protein against a mutated, DNA-attached, stressed protein confounds genotype, local DNA contacts, purification and mechanical load. Stronger controls keep attachment chemistry and sequence matched while altering the duplex's mechanical state (ssDNA, nicked/gapped dsDNA, intact dsDNA). The attached ubiquitin preprint in your project explicitly shows that local DNA–protein interactions can perturb conformational equilibria without the intended spring mechanism. These controls distinguish causal hypotheses, not merely improve sample quality.

18.4 Modeling fifteen HEAT repeats

A repeat-aware low-dimensional model could assign each repeat a rigid transform $T_j(q) = (R_j(q), \mathbf{t}_j(q))$, $j = 1, \ldots, 15$, and transform its reference atoms via $\mathbf{r}'_j(q) = R_j(q)\mathbf{r}_j + \mathbf{t}_j(q)$. Local observables include inter-repeat distances $d_{j,j+1}(q)$ and angles from Equation (60). A calibrated load-conditioned susceptibility is $\chi_j = \partial\langle\theta_j\rangle_F/\partial F$, with units rad/pN. This is a model-based coarse graining: HEAT repeats are not truly rigid, and maps need enough local resolution to support the fitted motions.

18.5 Where the previous representation tests inform model design

Your earlier SpringEM work highlighted a key risk: a 3DVA coordinate can fold back at large morph amplitudes. The right response is not to assume every nonlinear method automatically solves physics. A candidate representation must be tested for (i) injectivity/monotonicity with a known physical coordinate where one exists, (ii) pose robustness, (iii) held-out image agreement, (iv) density support from actual particles, (v) stable conditional populations across independent dataset splits, and (vi) sensitivity to the mechanical prior. An improved reconstruction error is not, by itself, evidence of accurate forces or free energies.

19 A complete synthetic reconstruction example: from pixels to a mechanical population

This section assembles the previously derived equations into one reproducible hypothetical experiment. All numerical values are illustrative, not measurements on PP2A. The point is to see the *types* and *operations* involved from beginning to end.

19.1 Step 1: acquisition and array shapes

Suppose we record 2,000 movies, each consisting of 40 detector frames of size 4096×4096. One movie has shape $4096 \times 4096 \times 40$. We correct motion and estimate a CTF per micrograph; each corrected micrograph has shape 4096×4096. Assume we pick $600,000$ candidate molecular particles over all micrographs. Extract boxes of $D = 256$ pixels at $a = 1.25$ Å/pixel, producing a particle tensor $I \in \mathbb{R}^{600000 \times 256 \times 256}$ (ignoring storage layout details). A 256-pixel box spans 320 Å. Every particle retains identifiers and optics metadata.

19.2 Step 2: particle cleaning and class probabilities

Suppose 2D classification yields 120 class averages, with each particle having class probabilities. We discard 40% as ice, contamination, overlaps or unsuitable picks, leaving $N = 360,000$. These choices must be audited because conformationally selective rejection biases populations. The subsequent reconstruction sees $360,000$ particle images and their metadata, not raw movies directly.

19.3 Step 3: estimate poses and a consensus map

Assume ab-initio reconstruction generates a plausible reference map. Refinement iterates pose likelihoods and volume updates until it produces a consensus V_0 on a 256^3 grid. We check FSC using independent half-maps and inspect directional/local resolution. If the protein changes shape substantially, the consensus is at best an average and some regions will be smeared.

19.4 Step 4: model heterogeneity

Suppose a low-rank method returns a one-dimensional coordinate for each particle. We build a density histogram but do not immediately call its negative logarithm free energy. We check whether the latent coordinate maps monotonically to physical attachment separation q, and whether pose errors or DNA-spring signal dominate the mode. For a calibrated physical q in nm, the density estimator must be normalized with units nm^{-1}.

19.5 Step 5: compare loading conditions

Suppose 180,000 accepted particles are relaxed controls and 180,000 intact-dsDNA chimeras. A two-state classification provides accepted counts:

Condition	State A	State B	Total
Relaxed	144,000	36,000	180,000
Hybridized	126,000	54,000	180,000

The observed proportions are $(0.8, 0.2)$ and $(0.7, 0.3)$. If these are unbiased equilibrium samples of matched constructs, the B/A odds change from $0.2/0.8 = 0.25$ to $0.3/0.7 \approx 0.42857$. The change in log odds is

$$\Delta \log \frac{P_B}{P_A} = \log\left(\frac{0.42857}{0.25}\right) = \log(12/7) \approx 0.5390.$$

Hence the relative loaded-versus-control bias favoring B is approximately $-0.5390\, k_B T$. This is a relative *state bias* under assumptions, not a force estimate. State probabilities must be corrected for misclassification and condition-dependent selection when these are present.

19.6 Step 6: propagate counting uncertainty

For a simple binomial model, estimated state-B proportion $\widehat{p} = n_B/n$ has standard error $\mathrm{SE}(\widehat{p}) \approx \sqrt{\widehat{p}(1-\widehat{p})/n}$. For relaxed control $\widehat{p} = 0.2$ with $n = 180000$,

$$\mathrm{SE}_{\mathrm{ctrl}} = \sqrt{0.2(0.8)/180000} \approx 0.000943.$$

For loaded $\widehat{p} = 0.3$,

$$\mathrm{SE}_{\mathrm{load}} = \sqrt{0.3(0.7)/180000} \approx 0.001080.$$

Their difference is 0.1 with simple independent-binomial standard error $\sqrt{0.000943^2 + 0.001080^2} \approx 0.001434$. This ridiculously small *nominal* uncertainty is misleading if particles are correlated within micrographs, class probabilities are uncertain, selected states differ in detectability, or preparation-level biological replication is insufficient. The real unit of independent replication may be grids, preparations or independent data collections, not individual particles.

Statistical pitfall: pseudoreplication. Reporting confidence intervals from hundreds of thousands of images as though those were independently prepared biological samples can dramatically overstate precision. Use independent preparation/grid-level replication and hierarchical models when assessing whether a spring changes conformational populations.

19.7 Step 7: what would make this evidence for mechanical coupling?

A defensible interpretation requires that (a) the two constructs are chemically matched apart from the mechanical state, (b) apparent populations are not explained by pose-dependent visibility or processing, (c) reconstructed states correspond to bona fide structural differences, (d) sample preparation plausibly preserves the relevant ensemble, and (e) an independent spring calibration or model supports the intended loading mechanism. If these conditions are satisfied, the population redistribution is evidence that the perturbation biases the structural ensemble. Establishing absolute force and free energy requires further calibration and equilibrium evidence.

20 Error propagation, posterior uncertainty and experimental replicates

20.1 Uncertainty is not a single number

Important uncertainty sources are detector noise, estimated motion, CTF parameters, particle picks, pose posterior, per-particle conformational uncertainty, map reconstruction, atomic-model fit, sample heterogeneity, and biological/preparation replication. Downstream analysis should not treat estimated latent values as exact truths. A full Bayesian approach would propagate a posterior over these coupled unknowns; practical approaches approximate it with independent half-datasets, bootstrapping, sensitivity analysis and hierarchical statistics.

20.2 Delta-method uncertainty for derived observables

If a physical observable is $q = f(\boldsymbol{\xi})$ and the input parameters $\boldsymbol{\xi}$ have covariance matrix Σ_ξ, then to first order

$$\mathrm{Var}(q) \approx (\nabla f)^{\mathsf{T}} \Sigma_\xi (\nabla f). \tag{75}$$

For 3D attachment positions $\mathbf{r}_A, \mathbf{r}_B$, define $d = \|\mathbf{r}_B - \mathbf{r}_A\|$ and unit direction $\widehat{\mathbf{e}} = (\mathbf{r}_B - \mathbf{r}_A)/d$. Its gradient with respect to $\mathbf{r}_A$ is $-\widehat{\mathbf{e}}$ and with respect to $\mathbf{r}_B$ is $+\widehat{\mathbf{e}}$. This permits approximate propagation of coordinate uncertainties into separation uncertainty, including correlations when known.

Worked example

If the two attachment positions each have independent isotropic coordinate uncertainty of standard deviation 0.2 nm per axis, then, to first order, the variance of their projected separation error is $0.2^2 + 0.2^2 = 0.08$ nm^2. The distance standard deviation is $\sqrt{0.08} \approx 0.283$ nm. Correlated shifts common to both atoms could cancel, so ignoring covariance may overestimate or underestimate uncertainty depending on its sign.

20.3 The law of total variance

If a particle's physical conformation q is uncertain because of the image I, the total variance of the ensemble decomposes as

$$\mathrm{Var}(q) = \mathbb{E}_I[\mathrm{Var}(q \mid I)] + \mathrm{Var}_I(\mathbb{E}[q \mid I]). \tag{76}$$

The first term is average posterior uncertainty *within* particles; the second is variability of per-particle posterior means *between* images. Keeping only the second term underestimates the total variance when within-particle uncertainty is appreciable. Conversely, a histogram of noisy point estimates can artificially inflate the apparent ensemble width.

20.4 Validation criteria for a proposed new analysis model

A new SpringEM-like method should be tested on synthetic data with known maps, poses, conformational coordinates and mechanical biases. Evaluate: reconstruction fidelity after common alignment; pose accuracy; recovery of physically ordered coordinates; calibration of predictive/posterior uncertainty; population estimation across conditions; robustness to picking/CTF/view bias; and held-out image likelihood. A well-performing model on synthetic images may still fail for experimental data when the assumptions differ, so include negative and matched-control tests.

21 A concise comparison with X-ray crystallography

Both cryo-EM and crystallography employ Fourier methods, but their measurement equations and missing-information problems differ fundamentally.

	Single-particle cryo-EM	Conventional X-ray crystallography
Fundamental measured data	2D electron images of projected particles	Diffraction pixel counts integrated to reflections
Dominant unknown geometry	Per-particle $SO(3)$ pose and image translation	Crystal lattice orientation/indexing
Fourier connection	Central-slice theorem	Structure factors at reciprocal-lattice points
Information challenge	Low-SNR unknown projections, CTF zeros, conformational variability	Missing diffraction phases and non-isomorphism
3D map	Electrostatic scattering-potential map	Electron density map
Ensemble analysis	Individual particle populations may provide heterogeneity information, subject to bias	Bragg structure primarily reports periodic average; diffuse/time-resolved data add further information
Perturbations	Matched stressed and relaxed grids or preparations	Perturbation-dependent diffraction or time-resolved exposures

The methods are complementary. In both, the measured quantities do not by themselves provide forces: mechanical conclusions depend on a structural coordinate, thermodynamic/statistical interpretation, controls, and independent calibration or a justified physical model.

22 A practical interpretation guide for your PP2A project

22.1 Which reconstruction is being compared?

For each experimental condition, keep the data provenance: original movies, corrected micrographs, particle extraction, classification criteria, CTF distributions, final half-maps, estimated poses and heterogeneous scores. Compare resulting maps at matched effective resolution and assess local confidence around the attachment points. Differences between WT-GST and Tet-modified DNA constructs cannot be attributed purely to force without matched-chemistry controls.

22.2 What would count as a useful structural result?

Even if the DNA spring does not cause a large global A-subunit displacement, a reproducible bias in specific HEAT-repeat angles or in terminal separation could be important. But distinguish whether it is observed in a reconstructed ensemble, inferred from a fitted structural model, or extrapolated from a neural interpolant. A map can support conformational changes without supporting precise atomic bond strains or force magnitudes.

22.3 What claims require independent information?

- **"There is structural heterogeneity."** Requires reconstruction evidence and validation against pose artifacts.
- **"The population shifts between conditions."** Requires accurate state assignment, sample-level replication, and checks against selection and orientation biases.
- **"DNA causes the shift through mechanical force."** Requires chemistry-matched, low-stress controls and a plausible spring-mechanics model.
- **"The applied force is F pN."** Requires independently calibrated geometry and a force–extension relationship.
- **"The energy landscape is $G(q)$."** Requires an equilibrium ensemble, a physically defined coordinate and measure, and justified treatment of selection biases.

23 Reference symbols and units

Symbol	Definition	Units
I_i	Extracted image of particle i	intensity
M_m	Motion-corrected micrograph indexed by m	intensity
$Y_m[u, v, n]$	Detector movie intensity at pixel and time/frame	counts / intensity
D, D_v	Side lengths of 2D image and 3D volume grids	pixels / voxels
a	Physical image sampling per pixel	Å/pixel
$V(\mathbf{x})$	Three-dimensional molecular potential map	potential
$\widehat{V}(\mathbf{K})$	3D Fourier transform of potential	transformed units
$\mathbf{k}, \mathbf{K}$	2D and 3D spatial-frequency coordinates	$Å^{-1}$
R_i	Particle 3D rotation matrix	dimensionless
$\mathbf{t}_i$	Image translation / shift	pixels or Å
$H_i(\mathbf{k})$	Per-particle contrast transfer function	dimensionless scale
λ	Electron wavelength in CTF formula	Å
Δf	Defocus in CTF formula	Å
C_s	Spherical-aberration coefficient	Å
A_i	Particle-specific forward linear imaging operator	linear operator
σ	Noise standard deviation	intensity
Σ	Noise covariance matrix	intensity^2
γ_{ik}	Posterior probability particle i belongs to class k	dimensionless
V_0	Consensus or canonical 3D map	potential
U_j	Signed variability mode shaped as a 3D map	model-dependent
a_{ij}	Component score for particle i and mode j	model-dependent
z_i	General learned conformational latent coordinate	arbitrary
q	Physically defined conformational coordinate	length / angle
$p(q)$	Probability density relative to dq	inverse units of q
$G(q)$	Potential of mean force along coordinate q	energy
$U_c(q)$	Applied perturbation free-energy bias in condition c	energy
k_B	Boltzmann constant	energy / K
T	Absolute temperature	K
β	Inverse thermal energy $1/(k_B T)$	energy^{-1}
F	Mechanical force conjugate to extension	pN
ℓ_p	DNA persistence length	nm
θ_j	Relative rotation angle between structural elements	radians

Symbol	Definition	Units
χ_j	Mechanical response of angle to force	radians / pN

24 Source notes and further reading

The mathematical derivations in this exposition are standard idealizations unless explicitly identified as a named software implementation. Numerical examples are synthetic teaching examples, not PP2A observations. Software-specific descriptions were checked against the following primary papers and guides (links included for direct access).

1. CryoSPARC Guide, *3D Variability Analysis tutorial, Part One*: CryoSPARC 3DVA guide (online). In particular, the guide specifies fixed consensus alignments, 3D covariance modes and linear representation limits.
2. CryoSPARC Guide, *3D Flexible Refinement tutorial*: CryoSPARC 3DFlex guide (online).
3. Zhong et al., *CryoDRGN: reconstruction of heterogeneous cryo-EM structures using neural networks*, Nature Methods (2021). DOI: 10.1038/s41592-020-01049-4.
4. Punjani and Fleet, *3DFlex: determining structure and motion of flexible proteins from cryo-EM*, Nature Methods (2023). DOI: 10.1038/s41592-023-01853-8.
5. Gilles and Singer, *Cryo-EM heterogeneity analysis using regularized covariance estimation and kernel regression*, PNAS (2025). DOI: 10.1073/pnas.2419140122.
6. Cheng et al., *A Primer to Single-Particle Cryo-Electron Microscopy*, Cell (2015). DOI: 10.1016/j.cell.2015.03.050. General imaging, alignment and resolution discussion.
7. Choi and Zocchi, *Guanylate kinase, induced fit, and the allosteric spring probe*, Biophysical Journal (2007). DOI: Choi–Zocchi DOI. DNA-spring mechanical perturbation context; user-supplied project PDF.
8. Tseng, Wang and Zocchi, *Enzyme-DNA chimeras: Construction, allostery, applications*, Methods in Enzymology (2021). DOI: 10.1016/bs.mie.2020.09.010. Spring construction, mechanics, and limitations; user-supplied project PDF.
9. Boral, Schnebly et al., *Conformational changes induced in ubiquitin by circular protein-DNA chimeras* (2026 preprint). DOI: 10.64898/2026.07.05.736590. Caution that DNA-induced local interactions can dominate the intended elastic effect; user-supplied project PDF.
10. Day et al., *Regulatory mechanisms of PP2A complex assembly driven by physicochemical differences in A-subunit isoforms*, Structure (2025). DOI: 10.1016/j.str.2025.06.013. PP2A A-subunit HEAT-repeat geometry and conformational variation; user-supplied project PDF.

Final statement. Cryo-EM is an inverse problem with latent pose, molecular structure and conformational variability. The leap from images to 3D maps requires Fourier geometry and statistical inference; the leap from maps to mechanics requires physically meaningful coordinates, equilibrium and selection assumptions, and calibrated interventions. Keeping these levels distinct is the prerequisite for a defensible force-resolved PP2A study.